Gesture Recognition

1. Experiment Objective

Use MediaPipe Hands combined with custom gesture classification logic to recognize 13 predefined gestures in real time (number gestures, special gestures, etc.).

2. Experiment Principle

1. Key Terms

Keyword	Description
13 gesture types	Zero, One, Two, Three, Four, Five, Six, Seven, Eight, Ok, Rock, Thumb_up, Thumb_down
Finger-up counting	Compares the y-coordinates of the fingertip (tip) and proximal interphalangeal joint (pip) to determine whether a finger is extended, returning a 5-bit binary array
Thumb-index distance	The key metric distinguishing the OK gesture from number gestures; a distance under 22% of the hand scale is judged as a pinch
Display scaling and black-bar padding	`resize_with_padding` scales the frame proportionally and pads it with black bars so the content stays complete and undistorted
Semi-transparent status overlay	A semi-transparent black bar at the top of the frame overlays the gesture name, FPS and operation hints

2. Technical Architecture

```
Terminal 1: start agent + camera + joint models
Terminal 2: start gesture recognition
```

3. Experiment Steps

1. Hardware Preparation

Hardware	Requirement
PTZ version	☑ Supported
No-PTZ version	☑ Supported
Required peripherals	ESP32 camera

2. Start the agent + camera + joint models (Terminal 1)

```
ros2 launch microros_v2_bringup car_base.launch.py
```

```
yahboom@yahboom:~$ ros2 launch microros_v2_bringup car_base.launch.py
[INFO] [launch]: All log files can be found below /home/yahboom/.ros/log/2026-07-28-10-47-55-372666-yahboom-149737
[INFO] [launch]: Default logging verbosity is set to INFO
[INFO] [micro_ros_agent-1]: process started with pid [149778]
[INFO] [camera_driver-2]: process started with pid [149780]
[INFO] [robot_state_publisher-3]: process started with pid [149782]
[INFO] [joint_state_publisher-4]: process started with pid [149784]
[INFO] [ekf_node-5]: process started with pid [149803]
[camera_driver-2] [INFO] [1785206876.886150022] [esp32_ip_camera_publisher]: camera node started, camera_url: http://192.168.12.37:81/stream
[joint_state_publisher-4] [INFO] [1785206877.007751174] [joint_state_publisher]: Waiting for robot_description to be published on the robot_descriptio
[micro_ros_agent-1] [1785206877.173080] info     | UDPv4AgentLinux.cpp | init                     | running...             | port: 8090
[robot_state_publisher-3] [INFO] [1785206877.754391885] [robot_state_publisher]: got segment base_footprint
[robot_state_publisher-3] [INFO] [1785206877.754890084] [robot_state_publisher]: got segment base_link
[robot_state_publisher-3] [INFO] [1785206877.754912957] [robot_state_publisher]: got segment bwheel_Link
[robot_state_publisher-3] [INFO] [1785206877.754918130] [robot_state_publisher]: got segment cam1_Link
[robot_state_publisher-3] [INFO] [1785206877.754924428] [robot_state_publisher]: got segment cam2_Link
[robot_state_publisher-3] [INFO] [1785206877.754929199] [robot_state_publisher]: got segment cam3_Link
[robot_state_publisher-3] [INFO] [1785206877.754935071] [robot_state_publisher]: got segment imu_frame
[robot_state_publisher-3] [INFO] [1785206877.754939792] [robot_state_publisher]: got segment laser_frame
[robot_state_publisher-3] [INFO] [1785206877.754944035] [robot_state_publisher]: got segment lfwheel_Link
[robot_state_publisher-3] [INFO] [1785206877.754949951] [robot_state_publisher]: got segment rfwheel_Link
[camera_driver-2] [WARN] [1785206879.871708745] [esp32_ip_camera_publisher]: Camera not opened, trying reconnect... (next retry in 1.0s)
[camera_driver-2] [INFO] [1785206880.046238399] [esp32_ip_camera_publisher]: IP camera stream opened successfully
```

3. Start gesture recognition (Terminal 2)

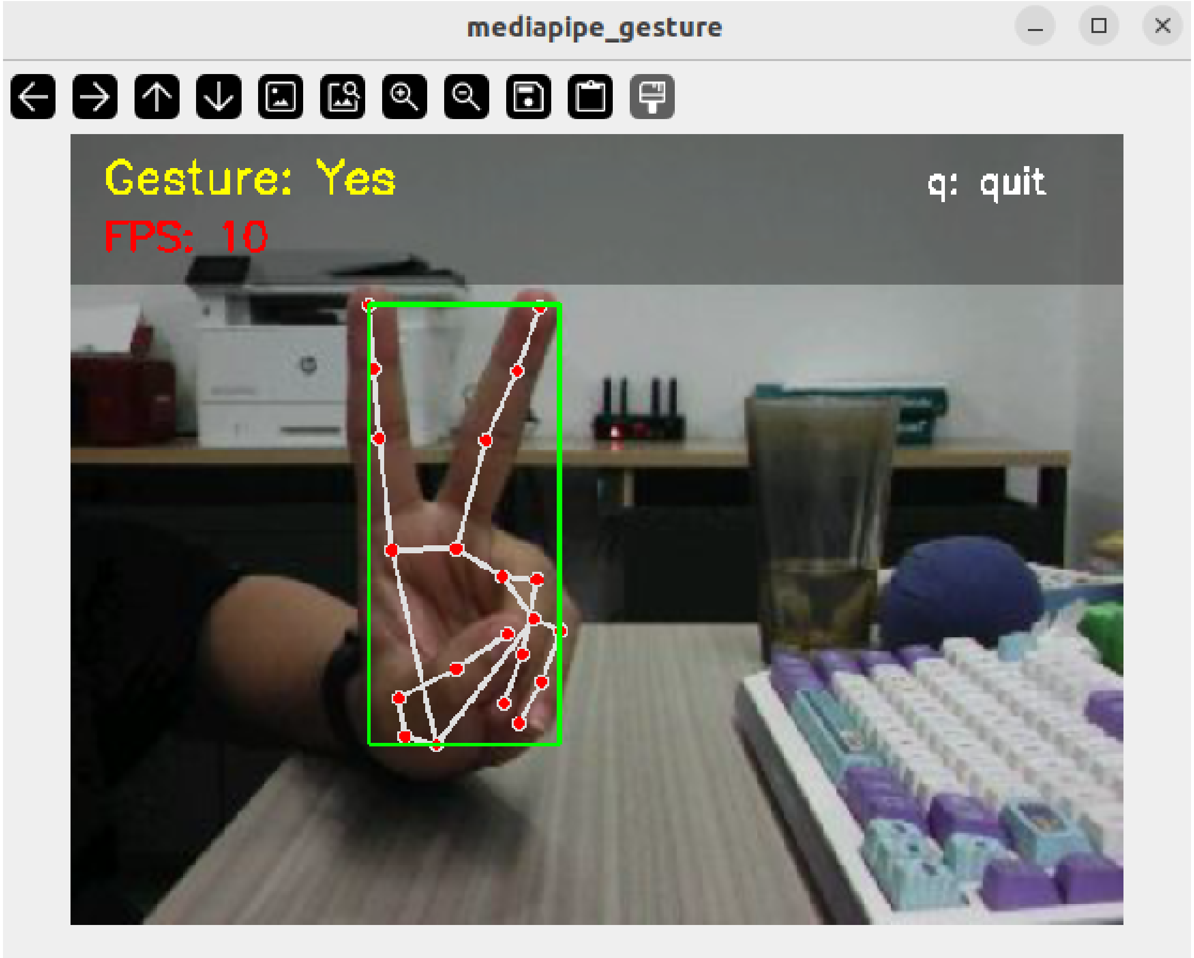
```
ros2 launch microros_v2_mediapipe_pkg mediapipe_gesture.launch.py
```

4. Operating Instructions

- Make one of the 13 gestures at the camera; the recognized result and FPS are shown at the top of the frame
- Press `q` to exit.

4. Observed Results

- Making number gestures (0-8) shows the corresponding gesture name at the top of the frame (e.g. "Five", "Ok")
- Finger extended/bent state is determined in real time through the `fingersUp` array
- A semi-transparent black overlay shows the gesture name, FPS and operation hints



5. Code Walkthrough

> Source code paths:
>
> - Launch: `~/microros_ws/src/microros_v2_mediapipe_pkg/launch/mediapipe_gesture.launch.py`
> - Node: `~/microros_ws/src/microros_v2_mediapipe_pkg/microros_v2_mediapipe_pkg/mediapipe_gesture.py`
> - Shared library: `~/microros_ws/src/microros_v2_mediapipe_pkg/microros_v2_mediapipe_pkg/media_library.py`

1. Core Node Analysis

```
# Source file:
~/microros_ws/src/microros_v2_mediapipe_pkg/microros_v2_mediapipe_pkg/mediapipe_gesture.py
class MediapipeGestureNode(Node):
    def __init__(self):
        super().__init__("mediapipe_gesture_node")
        self.declare_parameter("rgb_topic", "/camera/image_raw")
        self.declare_parameter("static_image_mode", False)
        self.declare_parameter("max_hands", 1)
        self.declare_parameter("detector_con", 0.7)
        self.declare_parameter("track_con", 0.7)
        self.declare_parameter("print_gesture", False)
```

```
        # Use the shared library HandDetector (wraps MediaPipe Hands + finger
state determination)
        self.hand_detector = HandDetector(
            static_image_mode=False, max_hands=1,
            detector_con=0.7, track_con=0.7)

    def image_callback(self, msg):
        frame = self.bridge.imgmsg_to_cv2(msg, 'bgr8')
        frame = self.hand_detector.findHands(frame)
        fingers_up = self.hand_detector.fingersUp()
        gesture = self.hand_detector.get_gesture(fingers_up)
        if gesture and self.print_gesture:
            self.get_logger().info(f'Gesture: {gesture}')
```

Key logic:

- `HandDetector` comes from the shared library `media_library.py`, encapsulating MediaPipe Hands and the gesture recognition logic
- `fingersUp()`: compares the fingertip y and pip joint y of the 5 fingers, returning a 5-bit array (1=extended, 0=bent)
- `get_gesture()`: classifies based on `fingersUp` + thumb-index distance into one of the 13 gestures
- `detector_con=0.7`: higher than the 0.5 used in basic hand detection, ensuring gesture recognition accuracy
- `print_gesture`: when enabled, the recognized gesture name is printed to the ROS2 log (for debugging)

2. Key Parameters

> Parameter definition files:
>
> - Launch: `~/microros_ws/src/microros_v2_mediapipe_pkg/launch/mediapipe_gesture.launch.py`
> - Node: `~/microros_ws/src/microros_v2_mediapipe_pkg/microros_v2_mediapipe_pkg/mediapipe_gesture.py`

Parameter	Type	Default	Description
`max_hands`	int	1	Maximum number of hands to detect
`detector_con`	float	0.7	Detection confidence threshold (higher than the base 0.5)
`track_con`	float	0.7	Tracking confidence threshold
`print_gesture`	bool	false	Whether to output the gesture name to the ROS2 log

3. Node Description

Node	Function
`mediapipe_gesture_node`	MediaPipe Hands + 13-gesture classification + semi-transparent overlay display

6. Frequently Asked Questions

Issue	Cause	Solution
Gesture misrecognized	Fingers are not fully extended or bent	Make the gesture properly, with clear extended/bent finger states
High recognition latency	`max_hands` > 1 increases computation	Set `max_hands:=1`